① (Most Relevant)

9. Translate a problem into a standard mathematical representation

✔ **Relevance: ★★★★★**

Analysis:

The key step is to construct the equation:

$$74 = 60 + 0.05x$$

👉 This represents a typical process of:

- Translating real-world language into an algebraic equation

Unlike qs1, this task emphasizes:

- **Reverse modeling (equation-solving structure)**

📌 Core competence:
Formulating a solvable mathematical equation from context

---

②

2. Identify the key variables in a model

✔ **Relevance: ★★★★★**

Analysis:

Key variable:

- x: number of newspapers sold (unknown)

Also identify:

- Fixed income: 60
- Unit rate: 0.05

👉 Incorrect identification of variables leads to failure in modeling

③

6. Identify and describe the mathematical aspects of a real-world problem

✔ **Relevance: ★★★★☆**

Analysis:

Students must abstract:

- Income structure = fixed component + linear growth component

👉 Extract the function:

$$f(x) = 60 + 0.05x$$

📌 This reflects:
Linear relationship modeling

④

1. Select a mathematical description

✔ **Relevance: ★★★★☆**

Analysis:

Students must choose:

- A linear function/equation representation

instead of:

- Proportional or piecewise models

👉 Reflects **model selection ability**

---

⑤

4. Read, decode and make sense of statements

✔ **Relevance: ★★★★☆**

Analysis:

Information is presented in a **poster-style format**, not directly as equations:

- "make 60 zeds a week"
- "plus 0.05 per newspaper"

👉 Students must correctly interpret:

- Fixed vs. variable components
- The additive structure ("plus")

---

⑥

7. Simplify or decompose a situation

✔ **Relevance: ★★★☆☆**

Analysis:

The situation is simplified into:

- Income = fixed + variable

👉 Compared to qs1 (piecewise structure), decomposition is simpler

⑦

10. Use mathematical tools to describe relationships

✔ **Relevance:** ★★★☆☆

Analysis:

Includes:

- Forming the equation
- Solving it using basic algebra

---

⑧

5. Recognise mathematical structure

✔ **Relevance:** ★★★☆☆

Analysis:

Recognizing:

- Linear function structure
- Constant rate of change

---

⑨

12. Identify constraints and assumptions

✔ **Relevance:** ★★☆☆☆

Analysis:

Implicit assumptions:

- Sales ≥ 0
- Income is fully determined by the given rule

⑩

8. Recognise similar known problems

✔ **Relevance: ★★☆☆☆**

Analysis:

Similar to:

- Base salary + commission models

However, transfer from prior knowledge is not strongly required

---

⑪

3. Select representation appropriate to context

✔ **Relevance: ★★☆☆☆**

Analysis:

The appropriate representation is:

- Algebraic expression (primary and sufficient)

---

⑫ (Least Relevant)

11. Use diagrams/symbols

✔ **Relevance: ★☆☆☆☆**

Analysis:

No graphical or diagram-based modeling is required

①（最核心）

9. Translate a problem into a standard mathematical representation

✔ **相关性：★★★★★**

分析：

该题关键在于建立方程：

$$74 = 60 + 0.05x$$

👉 这是一个典型的：

- 现实语言 → 代数方程转换过程

与 qs1 不同，这里强调：

- **逆向建模（equation solving structure）**

📌 核心能力：
将现实关系转化为“可求解方程”

---

②

2. Identify the key variables in a model

✔ **相关性：★★★★★**

分析：

关键变量：

- x：销售数量（未知量）

并识别：

- 固定收入：60
- 单位收益：0.05

👉 若变量识别错误（如把74当变量），模型无法建立

③

6. Identify and describe the mathematical aspects of a real-world problem

✔ **相关性：★★★★☆**

分析：

需要抽象出：

- 收入结构 = 固定项 + 线性增长项

👉 从现实情境提取出：

$$f(x) = 60 + 0.05x$$

📌 属于典型：
线性关系建模

④

1. Select a mathematical description

✔ **相关性：★★★★☆**

分析：

学生需要选择：

- 用“线性函数/方程”表示问题
  而不是：
- 比例或分段函数

👉 模型形式选择能力

---

⑤

4. Read, decode and make sense of statements

✔ **相关性：★★★★☆**

分析：

题干信息来自“广告海报”，表达方式非直接数学形式：

- “make 60 zeds a week”
- “plus 0.05 per newspaper”

👉 学生需正确理解：

- 固定项 vs 变化项
- “plus”的累加关系

---

⑥

7. Simplify or decompose a situation

✔ **相关性：★★★☆☆**

分析：

该题结构较简单，但仍需拆解：

- 收入 = 固定 + 可变

👉 相比 qs1（分段结构），复杂度较低

⑦

10. Use mathematical tools to describe relationships

✔ **相关性：★★★☆☆**

分析：

体现为：

- 建立方程
- 解方程（简单代数运算）

---

⑧

5. Recognise mathematical structure

✔ **相关性：★★★☆☆**

分析：

识别结构：

- 一次函数结构
- 线性增长关系

---

⑨

12. Identify constraints, assumptions

✔ **相关性：★★☆☆☆**

分析：

隐含假设：

- 销售量 ≥ 0
- 收入完全由该公式决定

⑩

8. Recognise similar known problems

✔ **相关性：** ★★☆☆☆

分析：

类似：

- 基础工资 + 提成模型

但迁移要求不强

---

⑪

3. Select representation appropriate to context

✔ **相关性：** ★★☆☆☆

分析：

主要使用：

- 代数表达式（唯一合理选择）

---

⑫（最低相关）

11. Use diagrams/symbols

✔ **相关性：** ★☆☆☆☆

分析：

无图形建模需求